DO NOW — RECALL

Lesson 144 · DNS and IP Routing: From Name to Destina

Name: _____ Class: _____ Date: _____

1. Which TCP/IP layer handles routing and IP addressing?

2. What data structure uses LIFO?

3. What is a trace table used for?

4. How many bits are in one byte?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Internet layer.

2. Model answer

Stack.

3. Model answer

Tracking variable state changes as an algorithm runs.

4. Model answer

Eight.

By the end, you can…

• Browser requests address for a website name through DNS.

• Application connects to a resolved IP address and transport port.

• Routers forward packets hop by hop to the destination network.

Vocabulary — know these terms

DNS — Hierarchical name-resolution system.

IP address — Network-layer address used to route packets.

resolver — Client/service that queries DNS and caches answers.

router — Forwards IP packets towards destination networks.

NEW LEARNING

Read, annotate and explain

Resolve hostname

Browser requests address for a website name through DNS.

Use IP destination

Application connects to a resolved IP address and transport port.

Route across networks

Routers forward packets hop by hop to the destination network.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

Name example.org

Identify the relevant information and make a prediction.

DNS returns an IP address

Explain the change using correct technical terms.

Connect to web server

DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain how DNS helps a user retrieve a website by name. [6]

Question 2

Differentiate DNS name resolution from IP packet routing. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A school website moves servers but some users reach the old address. Explain cached DNS and propose sensible checks. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

DNS sends the entire web page to the browser after it receives the hostname.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.